

1	Course Name:	Enterprise Networking																																										
	Course Code:	BMIT3084																																										
	Course Classification:	Major (core)																																										
2	Synopsis:	This course presents routing concepts, network security, WAN technologies, quality of service (QoS) mechanisms and network management concepts for enterprise networks.																																										
3	Name(s) of Academic Staff:	1	Refer to timetable																																									
		2																																										
		3																																										
4	Semester and Year offered:	Year Offered		Semester		Remarks:																																						
5	Credit Value:	4																																										
6	Pre-requisite/ co-requisite (if any):	NIL																																										
7	Course Learning Outcomes (CLO) 	CLO1	Employ routing concepts, network security, Access Control List (ACL), Dynamic Host Control Protocol (DHCP), Network Address Translation (NAT) for IPv4. (C3, PLO2)																																									
		CLO2	Examine Wide Area Network (WAN) concepts, Virtual Private Network (VPN) and IPSec, QoS Concepts. (C4,PLO2)																																									
		CLO3	Demonstrate TCP/IP Ethernet LANs and internetworks. (P4, PLO3)																																									
8	Mapping of the Course Learning Outcomes to the Programme Learning Outcomes, Teaching Methods and Assessment Methods																																											
	Course Learning Outcomes	Programme Learning Outcomes (PLO)											Teaching Methods	Assessment Methods																														
		PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10	PLO 11																																
	CLO1		✓											L,T,P,NF2F	Test, Examination																													
	CLO2		✓											L,T,P,NF2F	Examination																													
	CLO3			✓										L,T,P,NF2F	Practical Assessment																													
	Mapping with MQF Cluster of Learning Outcomes		C2	C3A																																								
Indicate the primary causal link between the CLO and PLO by ticking '✓' in the appropriate box. C1 = Knowledge & Understanding, C2 = Cognitive Skills, C3A = Practical Skills, C3B = Interpersonal Skills, C3C = Communication Skills, C3D = Digital Skills, C3E = Numeracy Skills, C3F = Leadership, Autonomy & Responsibility, C4A = Personal Skills, C4B = Entrepreneurial Skills, C5 = Ethics & Professionalism																																												
9	Transferable Skills (if applicable)																																											
	(Skills learned in the course of study which can be useful and utilized in other settings)		1	Cognitive skills																																								
			2																																									
			3																																									
		Open-ended response (if any)		4																																								
10	Distribution of Student Learning Time (SLT)																																											
	Note: This SLT calculation is designed for home grown programme only.																																											
	<table border="1"> <tr> <td colspan="10"></td> <td colspan="5">Learning and Teaching Activities**</td> </tr> <tr> <td colspan="10"></td> <td colspan="5">Face-to-Face (F2F)</td> </tr> </table>																									Learning and Teaching Activities**															Face-to-Face (F2F)			
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Course Content Outline and Subtopics		CLO*	Physical				Online/ Technology-mediated (Synchronous)				NF2F Independent Learning (Asynchronous)	Total SLT
			L	T	P	O	L	T	P	O		
1	Routing Concepts IP Static Routing and Troubleshooting	1,3	4	2	6						7	
2	OSPF • Single-Area OSPF • MultiArea OSPF	1,3	4	2	4						7	
3	Network Security Concepts • Malware and Common Network Attacks • IP Vulnerabilities and Threats • TCP and UDP Vulnerabilities • IP Services	1	2	1.5	0						6	
4	ACL Concepts ACLs for IPv4 Configuration	1,3	5	2	6						7	
5	NAT for IPv4 • NAT Characteristics • Types of NAT • Configure Static NAT • Configure Dynamic NAT • Configure PAT • NAT64 DHCPv4 Concepts SLAAC and DHCPv6 Concepts	1,3	4	1.5	6						6	
6	WAN Concepts • Purpose of WANs • WAN Operations • WAN Connectivity • Internet-Based Connectivity	2	2	1	0						5	
7	VPN and IPsec Concepts • VPN Technology • Types of VPNs • IPsec	2,3	2	1	2						4	
8	QoS Concepts • Network Transmission Quality • Traffic Characteristics • Queuing Algorithms • QoS Models • QoS Implementation Techniques	2	3	1.5	0						5	
9	Network Management • Device Discovery with CDP • Device Discovery with LLDP • NTP, SNMP, Syslog	2,3	2	1.5	4						5	
10												
11												
12												
13												
14												
15												
16												
17												
18												
19												
20												
SUB-TOTAL SLT:												122
Continous Assesement		%	Face-to-Face (F2F)				NF2F Independent Learning for Assessment (Asynchronous)					
			Physical		Online/ Technology-mediated (Synchronous)							
1	Test	24	2				8					
2	Practical Assessment	36	4				12					
3												
4												
5												
SUB-TOTAL SLT:												26
			Face-to-Face (F2F)				NF2F					

Final Assessment		%	Physical	Online/ Technology-mediated (Synchronous)	Independent Learning for Assessment (Asynchronous)
1	Examination	40	2		10
2					
3					
4					
5					
SUB-TOTAL SLT:					12
SLT for Assessment:					38
GRAND TOTAL SLT:					160
A	% SLT for F2F Physical Component: $\frac{[Total\ F2F\ Physical]}{[Total\ F2F\ Physical + Total\ F2F\ Online + Total\ Independent\ Learning]} \times 100$				48.75
B	% SLT for Online & Independent Learning Component: $\frac{[(Total\ F2F\ Online + Total\ Independent\ Learning)]}{[Total\ F2F\ Physical + Total\ F2F\ Online + Total\ Independent\ Learning]} \times 100$				51.25
C	% SLT for All Practical Component: $\frac{[%\ F2F\ Physical\ Practical + \%\ F2F\ Online\ Practical]}{100} \times 100$				17.50
C1	% SLT for F2F Physical Practical Component: $\frac{[Total\ F2F\ Physical\ Practical]}{[Total\ F2F\ Physical + Total\ F2F\ Online + Total\ Independent\ Learning]} \times 100$				17.50
C2	% SLT for F2F Online Practical Component: $\frac{[Total\ F2F\ Online\ Practical]}{[Total\ F2F\ Physical + Total\ F2F\ Online + Total\ Independent\ Learning]} \times 100$				0.00

Please tick (V) if this course is **Industrial Training/ Clinical Placement/ Practicum** using 50% of Effective Learning Time (ELT)

Note:

* Indicate the CLO based on the CLO's numbering in Item 8

** For ODL programme: Courses with mandatory practical requirements imposed by the programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

11	Identify special requirement or resources to deliver the course (e.g., software, nursery, computer lab, simulation room etc)	Computer Lab with networking equipment(Switch and Router) and networking simulation software (Packet Tracer)
12	References (include required and further readings, and should be the most current)	Main references supporting the course 1. Cisco Networking Academy. 2025. CCNA: Enterprise Networking, Security, and Automation. https://www.netacad.com
13	Other additional information (if applicable)	

Note: Number of PLO indicated is purely for illustration purposes only and the number is subjected to the curriculum design.